

SIDC — Scale-Invariant Dimensional Cascade

Gravity as Residual: A Thought Experiment on Dimensional
Inversion, Annihilation, and the Origin of the Dark Sector

v3.5.9+ A1 · 144 honest limitations · 421 pages · GitHub:
[ampbuster/gravity-as-residual](https://github.com/ampbuster/gravity-as-residual)

The Cone Structure

4D Bulk Event

$$E_{4D} = 5 \times 10^{79} \text{ J}$$

$$\tau_{4D} = 1.5 \times 10^{34} \text{ yr}$$

$$M_{Pl,4D} = 3.93 \times 10^{23} \text{ GeV}$$

Source: 4D-bulk energetic event



3+1D Universe (us)

$$M_{Pl,3D} = 1.22 \times 10^{19} \text{ GeV}$$

$$N_{\text{sub}} = 386 \text{ sibling sub-universes}$$

$$\text{Age} = 13.8 \text{ Gyr}$$

DE: 4D event's un-cancelled antigravity



2D Child Universes

$$M_{Pl,2D} = 2.95 \text{ TeV}$$

$$\alpha = 1.289, N=12$$

$$\tau_{2D}(\text{SN}) = 33 \text{ s}$$

DM: cumulative 2D universe deaths



Parameter Hierarchy (15 total, 4 truly new beyond SM)

#	Parameter	Value	Status
1	$M_{Pl,3D}$		MEASURED

#	Parameter	Value	Status
		1.22×10^{19} GeV	
2	$\alpha = 1 + 1/\sqrt{12}$	1.2887	FIRST-PRINCIPLES
3	$M_{\text{Pl},2\text{D}} = 12 \times v_{\text{H}}$	2.95 TeV	FIRST-PRINCIPLES
4	$\mu = M_{\text{Pl},2\text{D}}^2$	8.73×10^6 GeV ²	FIRST-PRINCIPLES
5	$N = 12$	12	FIRST-PRINCIPLES
6	$M_{\text{Pl},4\text{D}} = M_{\text{Pl},3\text{D}}^\alpha \times M_{\text{Pl},2\text{D}}^{1-\alpha}$	3.93×10^{23} GeV	DERIVED
7	$E_{4\text{D}} = N_{\text{sub}} \times E_{\text{sub}}$	5×10^{79} J	DERIVED
8	ε (bulk-brane coupling)	10^{-38}	CALIBRATED
9	$\tau_{4\text{D}}$ (4D event lifetime)	1.51×10^{34} yr	CALIBRATED
10	AGN rate (DM production)	3×10^{-16} /m ³ /s	CALIBRATED
11	$f_{\text{leak}} = c \times H_0$	2.47×10^{-18} /s	CALIBRATED
12	E_{sub}	1.30×10^{77} J	STRUCTURAL
13	$\tau_{3\text{D},\text{apparent}}$	8.95×10^{124} yr	STRUCTURAL
14	$\Upsilon_{4\text{D}}$	5.93×10^{90}	STRUCTURAL

#	Parameter	Value	Status
15	$N_{\text{sub}} \approx N_{12} \times (M_{\text{Pl},4\text{D}}/M_{\text{Pl},3\text{D}})^{1/3}$	386	PARTIAL

Truly new beyond SM+GR: 4 parameters ($M_{\text{Pl},3\text{D}}$, ρ_{DE} , AGN rate, N_{sub}). Comparable to ΛCDM 's 6, but with unified origin for DM and DE.



Recent Breakthroughs (v3.5.9+ A1)

L308ab: $f_{\text{leak}} = H(z)$ closes the CMB gap

User insight: "when the universe was small, pressure was higher, so more leaks back to 4d"

Generalized A1's $f_{\text{leak}} = H_0$ to $f_{\text{leak}}(z) = c \times H(z)$ with $c \approx 1.13$. This drains 32 orders of magnitude of overproduced DM by $z=1100$, matching Planck 2018 $\Omega_c = 0.265$. CMB gap was OPEN, now PARTIALLY CLOSED.

L308ad: $N_{\text{sub}} \approx N_{12} \times (M_{\text{Pl},4\text{D}}/M_{\text{Pl},3\text{D}})^{1/3} \approx 382$

First-principles formula for N_{sub} from $N=12$ and bulk/brane Planck ratio. Matches framework's 386 within 1.6% (within $M_{\text{Pl},4\text{D}}$'s $\pm 2\times$ uncertainty). L144 OPEN $\rightarrow$ PARTIAL.

L308ac: 4 truly new parameters beyond SM+GR

Of 15 listed parameters, only 4 are truly new: $M_{\text{Pl},3\text{D}}$ (gravity), ρ_{DE} (DE density), AGN rate (DM production), N_{sub} (now partial). 5 derive from SM, 4 are geometric, 1 is structural.



Test Status

PASS

Galaxies ($z < 4$)

11/11 specific cases pass real data. AGC 114905, KKR 25, DF2/DF4, Sun, etc.

PASS

RAR (SPARC, 175 galaxies)

MOND-compatible. 10% median residual. $g_+ \approx 1.2 \times 10^{-10} \text{ m/s}^2$ matches MOND a_0 .

PASS

CMB at $z=1100$ (L308ab)

$\Omega_c = 0.265$ matches Planck 2018. $f_{\text{leak}} = H(z)$ drains overproduction.

PASS

DE density

Matches within 0.13% via $f_{\text{DE}} \times \varepsilon \times M_{\text{PI},3\text{D}}^4$ closed loop.

PENDING 2027

47 Tucanae (DR1)

$M_{\text{dyn}} \approx M_{\text{stars}}$ (no local DM spike). Decisive test vs particle DM.

PENDING 2027-2028

DESI Y5

End-of-universe signatures from $\tau_{\text{DM}} = 14.5$ Gyr (now 12.84 Gyr with L308ab).

PENDING 2030s

SKA-MPG GW background

2D universe birth stochastic GW at 10^{60-62} erg/s/Mpc³ ($10^3\times$ below NANOGrav).

CONFOUNDED

Hubble tension (5.6 km/s/Mpc)

SIDC: $H_{0,4D} = 70.16$ (geometric mean of SH0ES+Planck).

Mechanism M: ACCEPT as real tension.



Version Timeline

- **v2.4** — Initial framework, $F_p(z)$ Hill function (DROPPED v3.3)
- **v2.7.5** — Smooth $F_p(z)$ closed CMB gap (later dropped for L308ab)
- **v3.0** — $N=12$ SYK Majorana breakthrough
- **v3.1.2** — Scenario X, brane-world, frame of reference explicit
- **v3.3** — Bilateral cascade ($F_p(z)$ dropped)
- **v3.5.7+** — $\mu = M_{Pl,2D}^2$ derived (5 paths)
- **v3.5.8** — $\alpha = 1+1/\sqrt{12}$ first-principles, MCMC 4/15 pinned
- **v3.5.9+ A1** — $f_{leak} = H_0$ as new principle (BREAKTHROUGH)
- **v3.5.9+ L308ab** — $f_{leak} = H(z)$ closes CMB gap (BREAKTHROUGH)
- **v3.5.9+ L308ac** — 4 truly new parameters beyond SM
- **v3.5.9+ L308ad** — $N_{sub} \approx N_{12} \times (M_{Pl,4D}/M_{Pl,3D})^{1/3}$ PARTIAL L144

Paper: [paper.pdf](#) (421 pages, 1.58 MB) · Source: github.com/ampbuster/gravity-as-residual

By a software developer (not a physicist) — thought experiment, not theory.